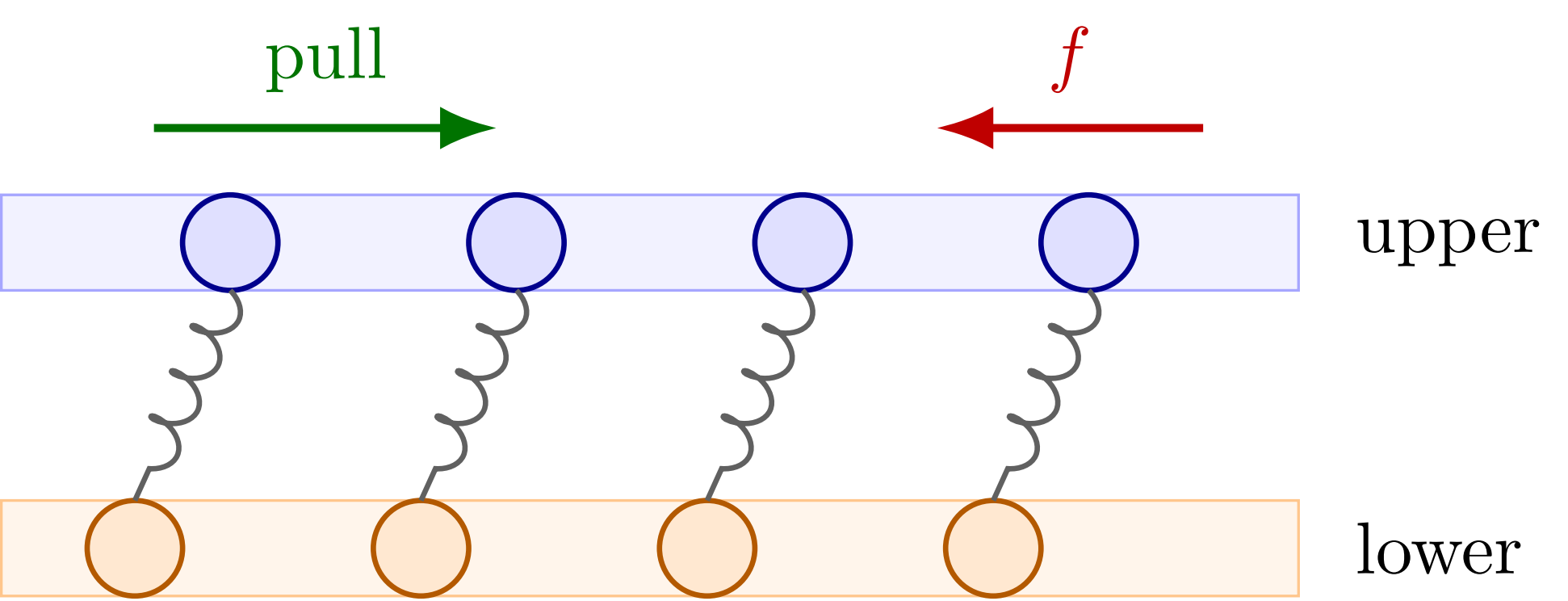
pull
$f$
upper
lower